

Does College Pay Off?

An Exploratory Data Analysis of Institutional Characteristics and Graduate
Earnings Using the U.S. College Scorecard

EDA Project Report

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Abstract

This report examines whether the cost, completion rate, student debt and type of a U.S. college are related to how much its graduates earn ten years after entering. Using the U.S. Department of Education's College Scorecard, 6,429 institutions were inspected and cleaned into an analytic sample of 1,938 four-year schools with complete data. Completion rate showed the strongest association with median ten-year earnings ($r = 0.51$), followed by in-state tuition and median debt (both $r = 0.47$). Institution type shifted the entire earnings distribution: private nonprofit schools had the highest median earnings (\$52,119), public schools were close behind (\$49,332) at roughly one quarter of the tuition, and private for-profit schools had the lowest earnings (\$40,092) despite charging twice the public tuition. The findings suggest that finishing a degree matters more for long-term earnings than how much the degree costs, and that public institutions often offer the strongest value. All results are associational and do not establish cause and effect.

1. Introduction

1.1 Background

A college degree is one of the largest financial decisions most Americans make. Published tuition at four-year schools ranges from a few thousand dollars a year at some public universities to nearly \$70,000 at the most expensive private colleges, and many students borrow to cover the difference. Families are routinely told that a degree is a good investment, but the price of that investment varies enormously from school to school. A natural question follows: do students who pay more, or who attend certain kinds of institutions, actually earn more later in life?

Answering that question requires data that follows students past graduation and into the workforce. Rankings and starting-salary surveys offer only a partial picture, because they rely on self-reported figures and on outcomes measured shortly after graduation. The College Scorecard, published by the U.S. Department of Education, fills this gap by linking each institution's cost, completion and borrowing records with federal tax data on former students' earnings up to ten years after they enrolled.

1.2 Purpose of the Study

The purpose of this exploratory data analysis (EDA) is to examine how the characteristics of a college relate to the long-term earnings of its students, and in doing so to shed light on the return on investment of different types of schools. Rather than testing a single formal hypothesis, the analysis looks for patterns, trends and relationships in the data that show which institutional factors are most closely tied to financial outcomes and where meaningful differences exist between institutions.

The dataset was chosen for three reasons. First, it comes from official federal records rather than surveys or opinions, which makes it reliable. Second, it combines costs, completion, debt and earnings for thousands of schools in one structured source. Third, it tracks earnings a full decade after entry, which captures real career outcomes rather than first-job salaries.

1.3 Research Questions

The analysis is guided by four questions:

- RQ1.** Is a school's completion rate related to the median earnings of its students ten years after entry?
- RQ2.** Do schools with higher tuition produce graduates with higher earnings?
- RQ3.** How is the median debt students carry at graduation related to their later earnings?

RQ4. How do cost, debt and earnings differ among public, private nonprofit and private for-profit institutions?

1.4 Initial Expectations

Going into the analysis, the expectation was that more expensive and more selective schools would generally show higher median earnings, but not uniformly, and that returns might flatten at the very highest tuition levels. It was also expected that reasonably priced schools would be found producing strong outcomes. Most importantly, the working hunch was that completion rate would be more strongly tied to earnings than tuition, suggesting that finishing a degree matters more than what it costs. These expectations were stated as three working hypotheses:

- **H1:** Schools with higher completion rates will have higher median earnings (a moderate positive relationship).
- **H2:** Higher tuition will be associated with higher earnings, but more weakly than completion rate, because tuition partly reflects prestige rather than directly causing better outcomes.
- **H3:** Median debt will show only a weak positive association with earnings, because debt levels are shaped largely by institution type and financial-aid structures.

1.5 Organization of the Report

Section 2 describes the data source, population and variables. Section 3 explains how the data were inspected, cleaned and analyzed. Section 4 presents the results, Section 5 discusses what they mean for each research question, Section 6 outlines the limitations of the study, and Section 7 concludes with recommendations for further work.

2. Data

2.1 Source

The data come from the College Scorecard “Most Recent Cohorts – Institution” file released by the U.S. Department of Education. The Scorecard combines several federal sources: the Integrated Postsecondary Education Data System (IPEDS) for institutional characteristics, the National Student Loan Data System (NSLDS) for federal aid and borrowing, and earnings records from the Internal Revenue Service matched to former students.

2.2 Population and Sample

The population represented is U.S. postsecondary institutions that participate in federal student aid programs, including public and private schools and two-year and four-year institutions. Each row

describes one institution and its aggregated outcomes, not an individual student. The raw file contains 6,429 institutions and 3,306 variables.

This is not a random sample. It is an administrative dataset that covers nearly all institutions meeting federal reporting requirements, so no sampling weights are needed. Two features of the data matter for interpretation. Institutions that do not take part in federal aid programs are excluded, and earnings and debt measures describe only students who received federal aid, which may differ from the full student body.

2.3 Variables

From the 3,306 available columns, eight were selected for analysis and five more were derived during cleaning and analysis. Table 1 serves as the data dictionary.

Table 1. *Data dictionary of variables used in the analysis*

Variable	Description	Type
INSTNM	Institution name	String
CONTROL	Institution type: 1 = Public, 2 = Private nonprofit, 3 = Private for-profit	Integer (categorical)
UGDS	Number of degree-seeking undergraduates enrolled	Float
TUITIONFEE_IN	In-state tuition and fees (\$)	Float
TUITIONFEE_OUT	Out-of-state tuition and fees (\$)	Float
C150_4	Share of first-time, full-time students completing within 150% of normal time at four-year schools	Float (0–1)
DEBT_MDN	Median debt at graduation; “PS” marks privacy-suppressed values	String
MD_EARN_WNE_P10	Median earnings of students 10 years after entry (\$)	Float
DEBT_MDN_NUM*	DEBT_MDN converted to numbers, suppressed values set to missing	Float
TUITION_GAP*	Out-of-state tuition minus in-state tuition (\$)	Float
ENROLL_BIN*	Enrollment size band: Small (<1k), Medium (1k–4.9k), Large (5k–14.9k), Very Large (15k+)	Category
CONTROL_LABEL*	Readable name for CONTROL	String
EARNINGS_TO_DEBT*	MD_EARN_WNE_P10 divided by DEBT_MDN_NUM	Float

* *Derived variable created during cleaning or analysis.*

3. Methodology

3.1 Data Inspection

Before cleaning, each selected field was inspected for data type, missing values, unusual codes and outliers. Missing data were common but largely expected. Completion rate was missing for 64.7% of institutions, because it applies only to four-year schools; tuition was missing for 42.0%, earnings for 17.9% and enrollment for 12.0%. Median debt was stored as text because 932 schools reported the privacy-suppression code “PS” instead of a number.

Boxplots and the interquartile range (IQR) rule identified high-end outliers in enrollment, tuition, debt and earnings. On inspection these were real institutions rather than data errors: very large online universities such as Southern New Hampshire University (156,755 undergraduates) and Western Governors University (135,822), high-tuition private colleges such as Kenyon College (\$69,330) and Columbia University (\$69,045), and high-earning technical and health schools such as MIT (\$143,372) and Samuel Merritt University (\$143,238).

3.2 Data Cleaning and Preparation

Cleaning followed the order below, with the data checked before and after each change. Table 2 shows how the number of institutions changed at each step that removed rows.

1. Only the eight analysis columns were kept; the remaining 3,298 columns were dropped as outside the scope of the research questions.
2. Schools above the 99th percentile on enrollment, in-state tuition or earnings were removed to limit the influence of extreme values on means and correlations.
3. Median debt was converted to a numeric field, with “PS” codes treated as missing.
4. Rows with impossible values (negative amounts or completion rates outside 0–1) were checked for; none were found.
5. Two new fields were created: the out-of-state tuition gap and enrollment size bands.
6. As the final step, rows missing enrollment, in-state tuition, completion rate or earnings were dropped. Missing values were removed last to avoid discarding usable data too early.
7. For the analysis, readable institution-type labels and an earnings-to-debt ratio were added.

Table 2. *Institutions remaining after each cleaning step*

Step	Institutions	Change
Raw College Scorecard file	6,429	—
After removing top 1% on enrollment, tuition or earnings	6,283	–146
After invalid-value check	6,283	0
After dropping rows missing key fields (analytic sample)	1,938	–4,345

The resulting analytic sample of 1,938 institutions includes 1,061 private nonprofit, 722 public and 155 private for-profit schools. By size, 642 are small, 870 medium, 313 large and 113 very large. Median debt remains missing for 69 of these schools, so analyses involving debt use 1,869 institutions.

3.3 Analytical Approach

Each research question was examined with simple, transparent methods suited to exploratory work. Descriptive statistics summarized the center and spread of each variable. For RQ1 to RQ3, the Pearson correlation coefficient (r) measured the strength and direction of the linear relationship between each factor and median ten-year earnings, and a scatterplot with a fitted regression line showed the pattern visually. For RQ4, group medians and boxplots compared institution types. Medians were preferred over means for group comparisons because several variables are skewed. The analysis was carried out in Python using pandas, Matplotlib and seaborn.

4. Results

4.1 Descriptive Statistics

Table 3 summarizes the main numeric variables for the analytic sample. Outcomes vary widely. The typical school's students earn about \$50,000 ten years after entry, but the middle half of schools spans roughly \$41,700 to \$59,000. Tuition is strongly right-skewed: the mean (\$22,664) sits well above the median (\$16,558) because a group of expensive private schools pulls the average up. On average, students earn about 3.8 times their median debt ten years after entry, though this ranges from below 1 to over 14.

Table 3. Summary statistics for the analytic sample

Variable	N	Mean	Median	Std. dev.	Min	Max
In-state tuition (\$)	1,938	22,664	16,558	16,376	600	64,910
Median debt (\$)	1,869	15,067	15,000	4,946	2,334	38,980
Median 10-yr earnings (\$)	1,938	51,135	49,840	13,845	17,360	95,951
Earnings-to-debt ratio	1,869	3.77	3.44	1.48	0.90	14.54

4.2 Completion Rate and Earnings (RQ1)

Completion rate showed the strongest relationship with earnings of any factor examined ($r = 0.51$, a moderate positive correlation). The fitted line rises from roughly \$30,000–\$35,000 at schools with very low completion to \$60,000–\$70,000 or more at schools where nearly all students finish. The

relationship is far from perfect, however. Schools with completion rates between 40% and 60% produce median earnings anywhere from about \$35,000 to over \$80,000.

Figure 1 colors each school by type and sizes it by tuition. Private nonprofit schools, many of them the larger high-tuition points, cluster in the upper right, where completion and earnings are both high. Public schools are spread widely and often reach \$45,000–\$65,000 at moderate completion rates. Private for-profit schools sit mostly in the lower left, with low completion and low earnings.



Figure 1. Completion rate vs. median earnings 10 years after entry, by institution type (point size = in-state tuition)

4.3 Tuition and Earnings (RQ2)

Higher in-state tuition was associated with higher earnings ($r = 0.47$), a moderate positive relationship somewhat weaker than that for completion rate. Figure 2 shows substantial scatter at every price level. Many low- and mid-tuition schools produce earnings equal to or higher than much more expensive schools, so tuition alone is an imperfect guide to outcomes.

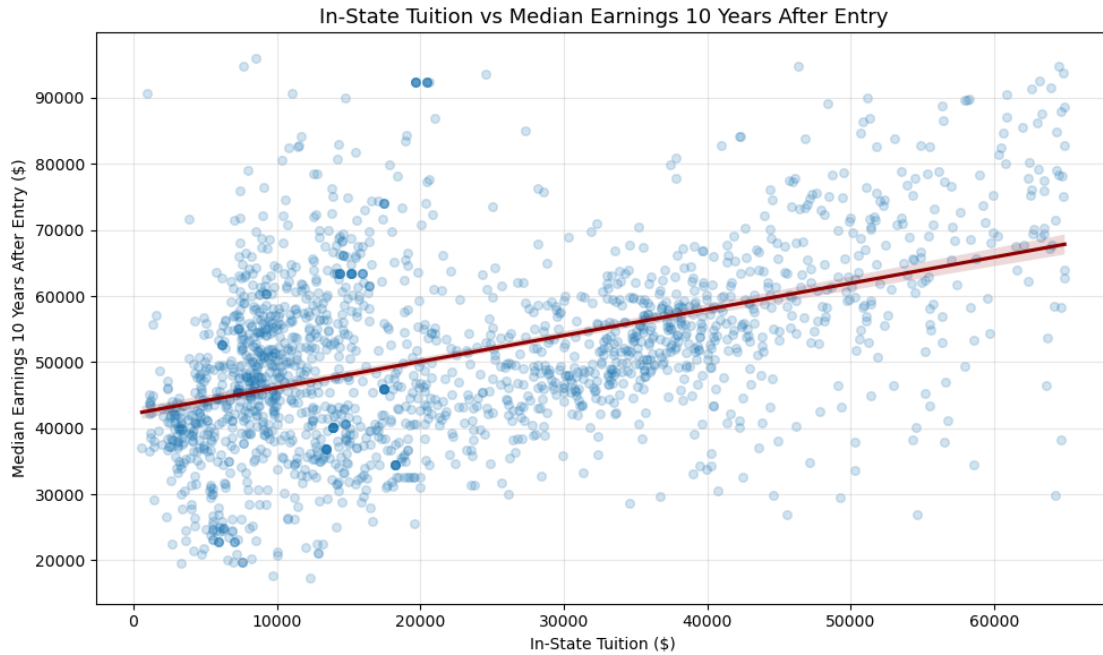


Figure 2. *In-state tuition vs. median earnings 10 years after entry*

4.4 Debt and Earnings (RQ3)

Median debt at graduation was also moderately and positively associated with earnings ($r = 0.47$), as shown in Figure 3. Schools with similar debt levels, particularly in the \$12,000–\$22,000 range where most schools fall, show a wide spread of earnings from low to very high. The earnings-to-debt ratio itself was only weakly related to earnings ($r = 0.16$).

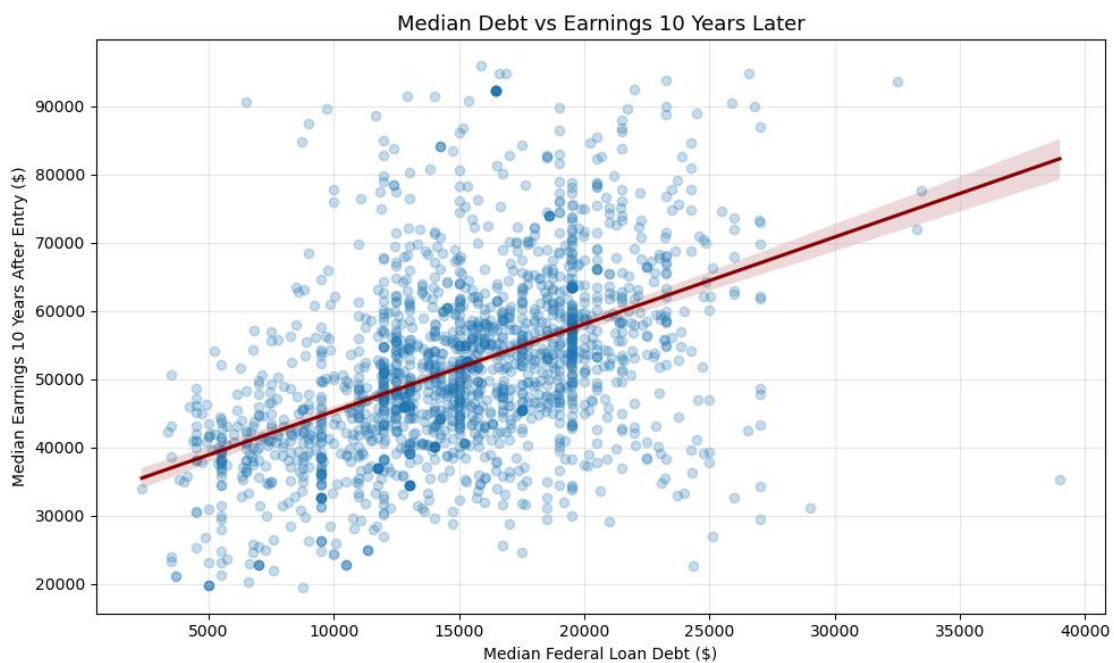


Figure 3. *Median federal loan debt vs. median earnings 10 years after entry*

4.5 Differences by Institution Type (RQ4)

Table 4 and Figure 4 compare the three institution types. Private nonprofit schools charge the most and report the highest median earnings. Public schools charge about one quarter as much in in-state tuition yet report earnings only about \$2,800 lower. Private for-profit schools charge roughly twice the public tuition but report the lowest earnings, about \$9,200 below public schools. Public schools also show the highest median earnings-to-debt ratio, about 4, compared with about 3 for the other two types.

Table 4. Median values by institution type

Institution type	Schools	In-state tuition	Median debt	10-yr earnings
Private nonprofit	1,061	\$33,494	\$17,074	\$52,119
Public	722	\$8,511	\$13,500	\$49,332
Private for-profit	155	\$17,488	\$13,000	\$40,092

The boxplot in Figure 4 shows that the three distributions overlap considerably, but the for-profit distribution sits clearly lower. Its middle half spans roughly \$35,000 to \$46,000, compared with about \$43,000 to \$57,000 for public schools and \$43,000 to \$60,000 for private nonprofits.

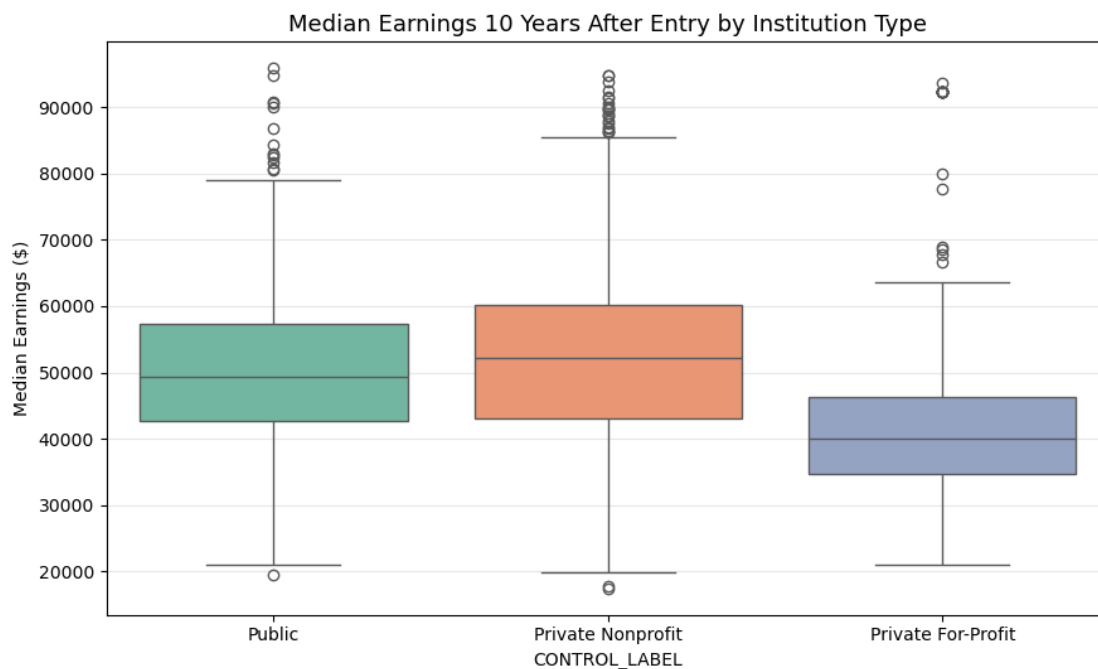


Figure 4. Distribution of median earnings 10 years after entry by institution type

4.6 Summary of Relationships

Table 5. *Correlation of each factor with median earnings 10 years after entry*

Factor	Pearson r	Strength
Completion rate (C150_4)	0.506	Moderate
In-state tuition (TUITIONFEE_IN)	0.468	Moderate
Median debt (DEBT_MDN_NUM)	0.468	Moderate
Earnings-to-debt ratio	0.156	Weak

5. Discussion

Completion rate (H1 supported). The results support the first hypothesis. Completion rate had the strongest association with earnings, and the pattern held within each institution type. One interpretation is that finishing a degree is what unlocks higher-paying careers, so schools that help more students finish produce better outcomes. Completion rate may also reflect other strengths, such as student preparation, advising and institutional resources.

Tuition (H2 largely supported). Tuition was positively associated with earnings, and, as expected, less strongly than completion rate. The gap between the two correlations was modest (0.51 versus 0.47), so tuition is not unimportant. However, the wide scatter at every price level shows that paying more does not guarantee better outcomes. Tuition most likely acts as a proxy for selectivity, program mix and institution type rather than causing higher earnings on its own.

Debt (H3 not supported as stated). Median debt was more strongly related to earnings than expected, matching tuition at $r = 0.47$ rather than showing only a weak link. This does not mean borrowing more leads to higher pay. Debt tends to be higher at more expensive private nonprofit schools, which also have higher earnings, so the relationship is likely driven by those shared factors.

Institution type. Institution type emerged as a major structural factor. Private nonprofits lead in absolute earnings, but public institutions deliver nearly the same earnings at a fraction of the cost, which suggests they often offer the best value. Private for-profit schools stand out negatively: they cost more than public schools yet have lower completion and lower earnings. For a given completion rate, the expected earnings level differs by type, which means institution type should be accounted for in any further modeling.

Taken together, the findings confirm the central hunch behind the study. What a student pays matters less for long-term earnings than whether they finish, and the kind of institution they attend shapes the level of outcomes they can expect.

6. Limitations

- **Association, not causation.** All results are correlations. They show that factors move together, not that one causes the other.
- **Omitted variables.** Important factors were not included, such as field-of-study mix (for example, STEM versus non-STEM), admissions selectivity, regional labor markets and student background.
- **Restricted population.** Earnings and debt measures describe only students who received federal aid, which may not represent all students at a school.
- **Sample restrictions.** Requiring complete data limited the sample to 1,938 four-year schools, so the results do not apply to two-year colleges. Removing the top 1% of schools on enrollment, tuition and earnings also excluded some of the most elite and highest-earning institutions, which limits what can be said about Ivy League and similar schools.
- **Diminishing returns untested.** The expectation that returns flatten at the highest tuition levels was not tested directly and remains an open question.

7. Conclusion and Future Work

This exploratory analysis of 1,938 U.S. four-year institutions found that completion rate is the clearest institutional signal of long-term earnings, that tuition and debt are moderately but inconsistently related to earnings, and that institution type shifts the level of outcomes students can expect. Public institutions stand out for delivering strong earnings at low cost, private nonprofits lead in absolute earnings, and private for-profit schools lag behind on nearly every measure. For students and families weighing the cost of college, the results suggest paying close attention to a school's completion rate and not assuming that a higher price brings a higher payoff.

Several extensions would strengthen these conclusions:

- Test whether the relationship between completion and earnings differs by institution type, using faceted plots or an interaction term in a regression model.
- Add selectivity measures such as admission rate (ADM_RATE) and average SAT score (SAT_AVG) to see how much of the gap between institution types they explain.
- Fit a multiple regression of earnings on completion rate, institution type, tuition and debt to estimate the separate contribution of each factor, checking for overlap between tuition and debt.
- Use the Scorecard's field-of-study file to measure how much earnings vary by major within the same school.
- Repeat the analysis without trimming the top 1% of schools to directly address the original question about elite and Ivy League institutions.

References

U.S. Department of Education. (2025). *College Scorecard: Most recent cohorts – institution data* [Data set]. https://ed-public-download.scorecard.network/downloads/Most-Recent-Cohorts-Institution_10032025.zip

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